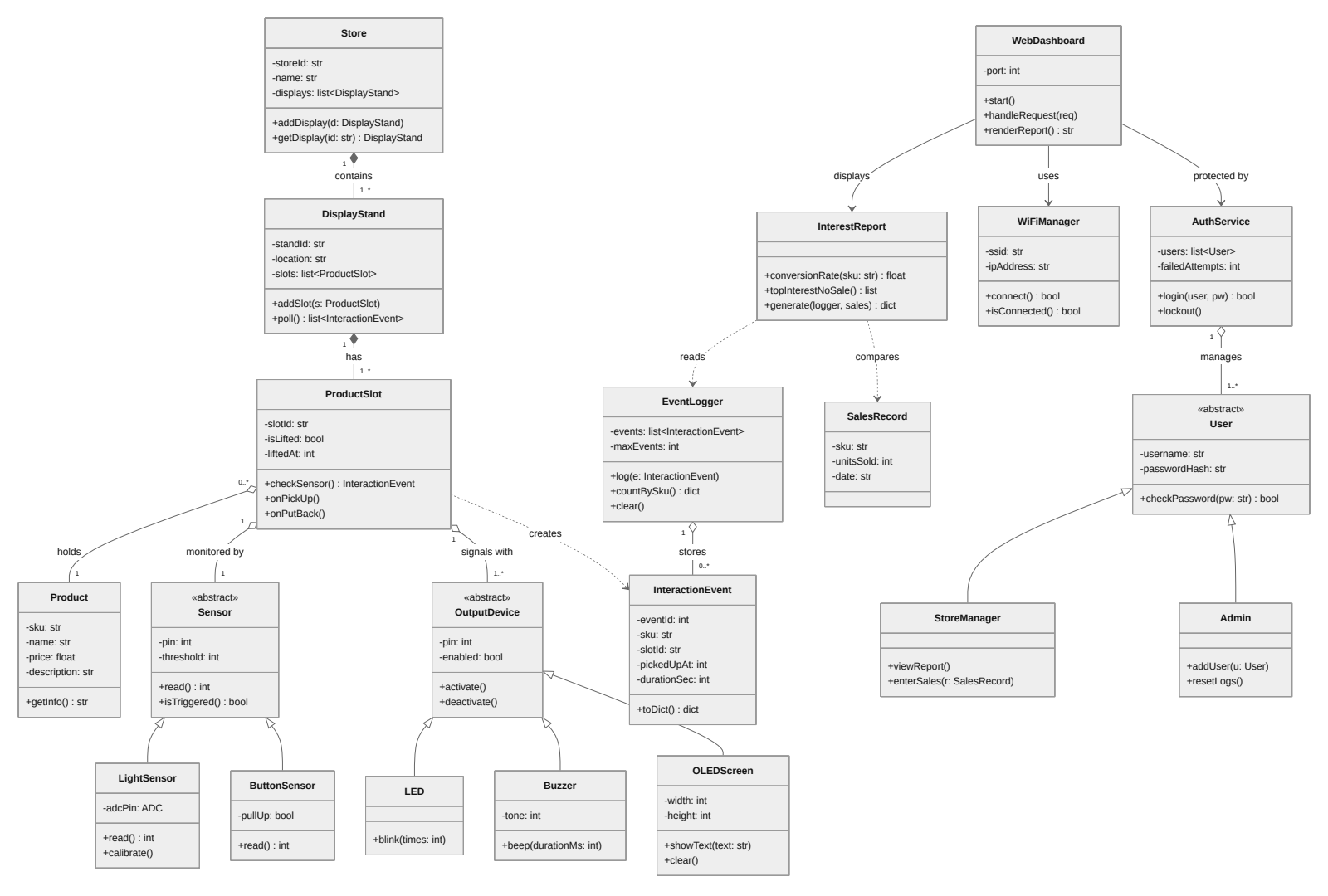

Store
-storeId: str
-name: str
-displays: list<DisplayStand>
+addDisplay(d: DisplayStand)
+getDisplay(id: str) : DisplayStand
1
contains
1..*
DisplayStand
-standId: str
-location: str
-slots: list<ProductSlot>
+addSlot(s: ProductSlot)
+poll() : list<InteractionEvent>
1
has
1..*
ProductSlot
-slotId: str
-isLifted: bool
-liftedAt: int
+checkSensor() : InteractionEvent
+onPickUp()
+onPutBack()
0..*
1
holds
1
Product
-sku: str
-name: str
-price: float
-description: str
+getInfo() : str
monitored by
1
«abstract»
Sensor
-pin: int
-threshold: int
+read() : int
+isTriggered() : bool
LightSensor
-adcPin: ADC
+read() : int
+calibrate()
ButtonSensor
-pullUp: bool
+read() : int
1
signals with
1..*
«abstract»
OutputDevice
-pin: int
-enabled: bool
+activate()
+deactivate()
LED
+blink(times: int)
Buzzer
-tone: int
+beep(durationMs: int)
creates
InteractionEvent
-eventId: int
-sku: str
-slotId: str
-pickedUpAt: int
-durationSec: int
+toDict() : dict
OLEDScreen
-width: int
-height: int
+showText(text: str)
+clear()
1
stores
0..*
EventLogger
-events: list<InteractionEvent>
-maxEvents: int
+log(e: InteractionEvent)
+countBySku() : dict
+clear()
reads
compares
InterestReport
+conversionRate(sku: str) : float
+topInterestNoSale() : list
+generate(logger, sales) : dict
SalesRecord
-sku: str
-unitsSold: int
-date: str
WebDashboard
-port: int
+start()
+handleRequest(req)
+renderReport() : str
displays
uses
protected by
WiFiManager
-ssid: str
-ipAddress: str
+connect() : bool
+isConnected() : bool
AuthService
-users: list<User>
-failedAttempts: int
+login(user, pw) : bool
+lockout()
1
manages
1..*
«abstract»
User
-username: str
-passwordHash: str
+checkPassword(pw: str) : bool
StoreManager
+viewReport()
+enterSales(r: SalesRecord)
Admin
+addUser(u: User)
+resetLogs()